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SECONDARY EMISSION COMPENSATION IN X-RAY SOURCES

Abstract

An X-ray imaging system is disclosed, comprising an X-ray source; a sample position; a detector arranged to detect X-ray radiation downstream of said sample position; wherein said X-ray source comprises an electron source arranged to provide an electron beam; a target arranged to produce X-ray radiation upon impact by said electron beam, the target comprising a substrate and a target layer at least partly covering said substrate, wherein said target layer is arranged to produce X-ray radiation upon impact by said electron beam; means for directing the electron beam to a first position on said target layer and a second position selected from a position on said target at which the electron beam impacts directly upon the substrate and a position on an electron beam dump arranged so that substantially no X-ray radiation created by interaction between the electron beam and the electron beam dump reaches the detector; a controller arranged to record, using said detector, a first image with the electron beam directed to said first position, and a second image with the electron beam directed to said second position, and generate a difference image between the first image and the second image. A method for X-ray imaging is also disclosed.

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Background/Summary

FIELD

[0001] The present disclosure relates to a method for X-ray imaging, and to a related X-ray imaging system.

BACKGROUND

[0002] It is well known in the field that secondary radiation can cause problems in X-ray sources, see e.g. US 2011/0176663. Secondary radiation is generally understood as any radiation other than the desired radiation and can originate, for example, from scattering of X-rays or from unintended electron beam interactions in the system. Typically, efforts are directed to reducing the amount of secondary radiation generated, which may be accomplished by mechanical design, e.g. to avoid direct line of sight from elements that may generate secondary radiation. Reduction of such secondary radiation may also be achieved by careful material selection, e.g. by providing elements that could otherwise generate secondary radiation with a coating of a low X-ray yield material (e.g. carbon). Furthermore, commonly used X-ray detector calibration procedures may, at least to some degree, reduce the influence of secondary radiation.

[0003] Further, a procedure known as flat field correction typically comprises exposing the detector to uniform radiation. Nominally this would result in equal readout from each pixel on the detector. However, various imperfections will result in variations between pixels. To compensate for this the values recorded during uniform illumination may be used to adjust subsequently recorded images. However, if the assumed uniform illumination comprises contribution from secondary radiation sources, i.e. it is not actually uniform, this procedure may reduce the influence of secondary radiation during imaging. A problem with this approach is that when a sample is present between the X-ray source and the detector the secondary radiation will, at least partly, be absorbed by the sample. Thus, since the assumption made of uniform illumination during the flat field calibration is not valid, the compensation may, when applied to an image comprising a sample, fail to eliminate the effects of secondary radiation.

[0004] A somewhat similar problem is related to scattered X-ray radiation making its way to the detector. This may particularly be an issue when doing CT scans. A way to compensate for this, proposed in EP3939510, is to calculate a simulated image created by the scattered radiation and subtract this simulated image from the acquired image to obtain a scatter reduced image. For this type of application, it is not evident how one should go about to actually obtain an image created only from the scattered radiation, hence a simulation was used.

[0005] In US 2005/0123097 there is disclosed a method where two images of the same object are acquired with different X-ray energy spectra. The two images are subtracted to enhance certain features, e.g. subtracting unwanted images of fatty tissue in a mammogram from images of potential cancer lesions or removing bone images from chest X-ray images. The two images are created by providing a multi-material target and directing the electron beam to different parts of the target to obtain different X-ray energy spectra. Embodiments where the different target materials

are provided as layers on top of each other, sometimes provided on a substrate transparent to X-rays, are also disclosed.

SUMMARY

[0006] The present disclosure proposes a way to compensate for undesired secondary X-ray radiation that may be created in an X-ray source by electron beam interaction with, for example, an aperture (a beam-limiting element for the electron beam) and/or a target substrate. Such secondary emission of X-ray radiation adversely affects the image quality that can be obtained when using the X-ray source for illumination. The proposed solution involves recording two images, one where the electron beam is directed to the target and one where the beam is directed in such a way that only the secondary radiation is emitted. The final image is then formed by generating a difference image between the two images.

[0007] In general, it would be preferable to eliminate all secondary radiation, i.e. in an ideal case all emitted X-ray radiation comes from interaction between the focused electron beam and the target at a desired location. However, in practice there will always be some secondary radiation generated elsewhere in the X-ray source. Secondary radiation may be viewed as having two components, affecting the generated X-ray images somewhat differently, as discussed below.

[0008] One type of secondary radiation is generated some distance upstream of the target (as seen along the electron beam). This type of secondary radiation will typically manifest as a well-defined spot in the final image. If a set of images are acquired with the object at different rotation angles and these images are used to make a three-dimensional reconstruction of the object, this type of secondary radiation may cause ring-like features in the reconstruction. In a sense, the image is created from two well-defined X-ray sources, one where the focused electron beam impacts the target and one where the electron beam interacts with some other structure upstream of the target. One way of isolating this type of secondary radiation is to blank out the electron beam, i.e. deflect the electron beam to a position away from the target where substantially no X-ray radiation is emitted from the X-ray source. Provided that the structure where the secondary radiation is created is located upstream of the beam deflector, the contribution to the final image from the secondary radiation will be the same when directing the electron beam towards the target as when the beam is blanked, and the effect thereof in the final image can be eliminated by generating a difference image between two images (blanked and non-blanked electron beam).

[0009] Another type of secondary radiation is generated close to the target, e.g. by interaction between the electron beam and a target substrate or by interaction between electrons scattered from target and a backscatter collector. This type of radiation will typically manifest as a general blurring of the resulting X-ray image. Since the secondary radiation is generated close to the exit window it may be emitted over a large angular range, thereby in a sense making the effective X-ray spot larger and less well defined. To be able to isolate the contribution from this type of secondary radiation, the electron beam must impact the target and/or the target substrate under conditions such that the X-ray radiation created by interaction between the electron beam and the target is not emitted from the source.

[0010] One example of a source of secondary radiation is apertures or other beam-limiting elements in the electron beam path. The purpose of having an aperture that geometrically limits the electron beam is to have a well-defined electron beam size at some point and/or to limit the angles at which electrons propagate further in the system. The dimension of the aperture is determined inter alia by requirements on electron beam spot size at the target and the total electron beam throughput. However, electrons not passing through the aperture will interact with the material defining the aperture (i.e. with the beam-limiting element) and at least to some extent cause X-ray radiation to be generated. To avoid this, the material defining the aperture may be made of, or at least be coated with, a material with low X-ray generation efficiency. An alternative may be to provide the aperture as a hole in a relatively thick and dense material, such that X-ray radiation generated by electrons impacting on this material is absorbed (e.g. reabsorbed in the material) and

does not reach the exit window. It may be difficult to eliminate the secondary radiation entirely using these approaches, but at least some reduction may be achieved. Another approach may be to not have a direct line of sight between the aperture and the exit window through which X-ray radiation is emitted from the source. The latter option may require some added complexity in some cases, e.g. in transmission configurations where X-rays are by definition emitted in the same direction as the electron beam. Furthermore, downstream from the aperture there should not be any other objects than the target with which the electron beam can interact to generate X-ray radiation that may exit the source.

[0011] A further potential source of secondary radiation is interaction between electrons backscattered from the target and surrounding parts of the X-ray source. It is known in the art to provide a backscatter collector (or trap) to capture the scattered electrons and the energy they carry. Interaction between the backscatter collector or some other part of the X-ray source and the electrons may in turn generate secondary radiation that can be emitted through the exit window. To eliminate, or at least reduce, this contribution to the emitted radiation, surface treatments (see WO 2020/052773) and/or geometric constraints (see US 2012/0170715) may be applied.

[0012] In cases where the X-ray source comprises electron optics for focusing and deflecting the electron beam, and in cases wherein the target comprises parts where a low X-ray yield substrate is exposed, the inventive principles disclosed herein may be implemented by taking a first image with the electron beam directed to an intended working position, i.e. on a target layer for instance comprising tungsten (W), and taking a second image with the electron beam directed to a location where the substrate, e.g. comprising diamond, is exposed (i.e. where there is no target layer on the substrate). By moving the sample so that it is imaged at the same location on the detector for each image, the image generated by X-rays emanating from the target layer may be extracted by generating a difference image between the images (e.g. by subtracting corresponding pixel values of the two images). Provided the shift in image position on the detector is small compared to the pixel resolution the sample may not need to be moved. Secondary radiation generated from some other structure, e.g. material defining an aperture, in the X-ray source will be present in both images and thus eliminated when generating the difference image.

[0013] Provided that electron beam steering and shaping is performed downstream (relative to the electron beam propagation direction) from the structures generating the secondary radiation, the contribution from this secondary radiation will be the same in the two images.

[0014] In other embodiments, the electron beam may be directed in the same way when recording the first and second images while instead the target is moved between image acquisitions so that the first image is exposed with the electron beam directed to the target layer and the second image is recorded with the electron beam directed so that only secondary radiation contributes to the image. In this way there is no need to move the sample between the images since the relative orientation between electron beam, sample, and detector is the same for the two images. The X-ray source in this case comprises an actuator arranged to move the target with sufficient precision, typically the order of μm , in a vacuum tight arrangement.

[0015] X-ray radiation generated by electrons penetrating the target layer, i.e. X-ray radiation generated from electron beam interaction with the substrate supporting the target layer rather than the target layer itself, will create an image that is substantially the same as the one generated by electrons impacting directly on the substrate, i.e. on a bare portion of the substrate, provided that scattering from the target layer may be neglected.

[0016] In cases where negligible secondary radiation is generated by interactions between the electron beam and structures located upstream beam deflection elements a first image may be acquired with the electron beam directed towards the target layer and a second image may be acquired with the electron beam directed towards an exposed surface of the target substrate. To compensate for a shift of image position on the detector the sample may be moved accordingly, or the acquired images may be shifted (aligned) so that they overlap prior to creating a difference

image. Since there is negligible secondary radiation that is independent on where the electron beam impacts on the target no noticeable artefacts will be created by aligning the images.

[0017] In cases where a substantial fraction of the incoming electrons is absorbed and/or scattered in the target layer, a more elaborate procedure may be required. By deflecting the electron beam to a location from which no generated X-ray radiation can exit the X-ray source or at least not reach the detector, e.g. a shielded electron dump, an image created only from secondary radiation generated from other structures (e.g. the beam-limiting element for the electron beam) within the X-ray source may be obtained. The electron dump may be arranged as any part of the source that can withstand the thermal load imposed by the electron beam and wherein any X-ray radiation generated by the impacting electron beam does not reach the detector. By generating a difference image (e.g. by subtracting pixel values of this image from corresponding pixel values of the image acquired when the electron beam is directed towards the target layer), the influence of the secondary radiation generated at the beam limiting element may be eliminated or at least substantially reduced. To also eliminate the influence of X-ray radiation generated in the substrate, an image may be acquired with the electron beam directed towards the substrate but away from the target layer and with the sample moved so that it is imaged at the same position on the detector as when the electron beam is directed towards the target layer. The contribution from secondary radiation generated at the beam-limiting element may be eliminated also from this image by subtraction of the image acquired with the electron beam directed to the electron dump. The resulting image may then be scaled to compensate for absorption and/or scattering in the target layer. After scaling, pixel values of the image may be subtracted from corresponding pixel values of the difference image generated between the image recorded with the electron beam directed towards the target layer and the image generated with the electron beam directed to the electron dump. The scale factor may be known from target design or may be determined during a calibration procedure. As will be understood, a suitable target thus comprises a substrate that is only partly covered with a target layer (e.g. made from tungsten), while some parts of the substrate lack such target layer.

[0018] In the discussion above, it has been assumed that the images associated with secondary radiation generated at the aperture (beam-limiting element) may be considered identical irrespective of sample movement. In most practical cases this will be an acceptable assumption. In case it is not, i.e. when an image of the sample created by the secondary radiation changes considerably upon sample movement, a more elaborate method may be employed. A total of four images may be recorded. A first image is then taken with the sample in a first position and with the electron beam directed towards the target. A second image is taken with the sample still in the first position and with the electron beam directed so that only the secondary radiation from the aperture reaches the detector. A third image is taken with the electron beam deflected towards a part of the target where the substrate is exposed, and the sample is moved to a second position so that the image of the sample is at the same location as for the first image. A fourth image is taken with the sample in the second position and with the electron beam directed as for the second image. By subtracting the second image from the first image to generate a fifth image and subtracting the fourth image from third image to generate a sixth image, respectively, the effect from secondary radiation generated at the aperture is eliminated (or at least reduced). Then, by subtracting the sixth image, suitably scaled to compensate for scattering and/or absorption in the target layer, from the fifth image the effect from secondary radiation generated by electrons interacting with the substrate is also eliminated. As discussed above, subtracting an image from another means that a difference image between the two images is generated, e.g. by subtracting pixel values of one image from corresponding pixel values of the other image. For example, a difference image between the first and the second image can be generated by subtracting pixel values of the second image from corresponding pixel values of the first image. Pixel values may in some embodiments be scaled prior to subtraction.

[0019] For X-ray sources where secondary radiation created by electrons backscattered from the target make a substantial contribution to the image formed on the detector it may be advantageous to provide regions on the target for which no X-ray radiation is emitted towards the sample and the detector when the electron beam is directed thereto. This may be accomplished by providing local X-ray shielding, e.g. on the outside of the target for a transmission target, or by providing a region of the target that has no line of sight to the exit window. By directing the electron beam to such a location, only X-ray radiation created by the backscattered electrons will reach the detector, thus forming an image that can be subtracted from an image acquired with the electron beam directed to the intended target location.

[0020] A corresponding procedure may be performed to eliminate contributions from backscattered electrons to images formed with the electron beam directed to the exposed substrate of a transmission target. Once the backscatter contribution has been eliminated the image acquired with the electron beam directed towards the substrate may, possibly after suitable scaling, be subtracted from the image acquired with the electron beam directed towards the target layer.

[0021] In embodiments where an image comprising substantially only X-ray radiation generated at some structure upstream of the electron beam deflector (e.g. at an aperture) the image may be thresholded before subtraction from another image. The thresholding may be performed so that only pixels that actually received secondary radiation are used in the subtraction while other pixel values are set to zero (i.e. in effect the corresponding pixels in the original image are left unchanged after subtraction). A suitable threshold may for example be found as a multiple of the standard deviation of the pixel values or as a fraction of the maximum pixel value in the image. This may be an advantageous procedure in that the amount of noise in the final image may be less than what would be the case if all pixel values were used when generating the difference image (i.e. in the subtraction procedure). Other means for reducing the noise may also be employed, such as local median filtering of pixel values or other well-known image processing algorithms.

[0022] The methods outlined above may be applied when doing CT scans. In particular, images with the electron beam blanked out may be recorded for at least some of the sample orientations used during the scan to compensate for secondary radiation generated at an aperture, as discussed above. It may be advantageous to acquire the compensation images for a subset of the sample orientations used for the entire scan, for example good reconstruction results have been obtained by acquiring a blanked-out image for every tenth sample orientation. Instead of blanking out the electron beam it may be directed to a part of the target where the substrate is exposed, in this way somewhat sharper images may be obtained after subtraction of the compensation images. However, by blanking out the electron beam the total X-ray dose applied to the sample may be smaller.

[0023] The different approaches described above may be combined in different ways depending on which contributions to the secondary emission that need to be considered.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] In the detailed description below, reference will be made to the accompanying drawings, on which:

[0025] FIG. 1 is a schematic overview of an X-ray system according to the invention.

[0026] FIG. 2 illustrates how the electron beam may be directed to different locations on the target and how the sample may be moved accordingly.

[0027] FIG. 3 illustrates schematically a method for X-ray imaging.

DETAILED DESCRIPTION

[0028] While the inventive ideas presented herein have been discussed and summarized above to provide a thorough understanding thereof, specific implementations will be described below to

further illustrate how the invention can be put into practice.

[0029] By way of introduction, FIG. 1 schematically shows an exemplary imaging system comprising an X-ray source **100**. The X-ray source as shown includes an electron gun **102** configured to form an electron beam **114**. The electron beam emitted from the electron gun **102** is aligned and shaped using alignment coils **104** and stigmator coils **106**, respectively, before reaching a beam-limiting element in the form of an aperture **108**. As discussed in the summary above, such an aperture has the purpose of limiting the geometrical spread of the electron beam before it enters downstream (relative to the propagation direction of the electron beam) electron beam optics. The aperture **108** typically has a diameter, i.e. opening dimension, in the order of 1 mm. After having passed the aperture **108**, the electron beam reaches electron optics, typically comprising one or more focusing lenses **110** and one or more deflectors **112**. The electron beam optics is used for directing and focusing the electron beam to an intended location on a target **116**. The target **116** typically comprises a high-Z ($Z > 20$) material that generates X-rays **118** upon electron impact. A typical target material in this type of X-ray source is tungsten (W), present as a film on a substrate material, e.g. made from diamond. These components of the X-ray source are enclosed within a low-pressure (vacuum) enclosure **120**. The generated X-ray radiation **118** may be used for imaging a sample or an object **130**, located at a sample position, using a detector **140**. The X-ray source also includes a controller **150** that, according to the principles disclosed herein, is configured to record, using the detector **140**, a first image with the electron beam directed to a first position, record a second image with the electron beam directed to a second position, and generate a difference image between the first and the second image, e.g. by subtracting pixel values of the second image from corresponding pixel values of the first image. The controller **150** may also be responsible for controlling the lenses **110** and the deflector **112** to direct the electron beam to said positions. As illustrated in FIG. 1, the controller **150** may be operationally coupled to the electron optics **110**, **112** and the detector **140**.

[0030] Hence, an X-ray imaging system according to an embodiment comprises an X-ray source **100** as discussed above, a sample position (illustrated by the sample **130** in FIG. 1), and a detector **140**. The X-ray source **100** comprises an electron source **102** arranged to provide an electron beam, a target **116** arranged to produce X-ray radiation upon impact by the electron beam, means such as electron optics **110**, **112** for directing the electron beam to a first position on the target and a second position (which may or may not be on the target), and a controller **150** arranged to record, using the detector, a first image with the electron beam directed to the first position, record a second image with the electron beam directed to the second position, and generate a difference image between the first and the second image. The imaging system may also comprise an actuator (not shown) arranged to move the target in relation to the electron beam, so that directing the electron beam to the first and second positions may involve steering/deflecting the electron beam using electron optics and/or moving the target. The imaging system may also comprise a manipulator for moving the X-ray source and the sample position in relation to each other. Such manipulator may, for example, be a translation stage **135** to which a sample can be mounted, as shown in FIG. 1; other examples include putting the X-ray source on a translation stage or providing the X-ray source with feet that enable translation. Preferably, the manipulator is also operatively connected to the controller **150**, as schematically illustrated by a dashed line for the translation stage **135** in FIG. 1.

[0031] FIG. 2 schematically shows the target **210**, **220**, the sample/object **230**, and the detector **240** in more detail. According to the principles disclosed herein, influence of unwanted X-ray radiation on the captured image is reduced by subtracting one or more reference images from a main image. The unwanted X-ray radiation is generally referred herein to as secondary emission radiation. In order to capture the main image and the one or more reference images, the electron beam is moved (deflected) between different locations for capturing the various images involved. In FIG. 2, three different electron beam locations are illustrated; a first location is shown at **214a**, where the electron beam impacts a target layer **220** for intended generation of X-ray radiation; a second

location is shown at **214b**, where the electron beam does not impact the target layer **220** but instead directly impacts the underlying substrate **210**; a third location is shown at **214c**, where the electron beam impacts an electron dump **250**. As will be understood, arrows **214a**, **214b** and **214c** in FIG. 2 represent the same electron beam directed at different locations. When the electron beam impacts directly upon the substrate **210**, as shown at **214b**, some X-ray radiation may be generated by interaction between the electron beam and the substrate material. It will be appreciated that X-ray radiation may also be generated by interaction between the electron beam and the substrate when the electron beam is directed to the target material **220**, as shown at **214a**, since some electrons may pass through the target material **220**. On the other hand, when the electron beam impacts the electron dump **250**, no X-ray radiation generated from interaction between the electron beam and the electron dump exits the X-ray source, or at least no X-ray radiation generated from interaction between the electron beam and the electron dump reaches the detector. Hence, when the electron beam impacts the electron dump **250**, any X-ray radiation reaching the object **230** and the detector **240** is predominantly comprised of X-ray radiation generated at the aperture **108** located upstream from the target (see FIG. 1).

[0032] Regardless of whether the electron beam is directed to location **214a**, **214b** or **214c**, a background of X-ray radiation generated at the aperture **108** will be present at the object **230** to be imaged. Such background can be reduced in the captured image by taking the difference between an image captured with the electron beam at location **214a** and an image captured with the electron beam at location **214c**. Taking a difference between images may for example involve taking a difference between each corresponding pixel values of the images.

[0033] When the electron beam is directed towards the substrate **210**, as shown at **214b** in FIG. 2, the X-ray radiation reaching the object **230** may include both a contribution from radiation generated at the aperture **108** and a contribution generated in the substrate **210**.

[0034] When an image captured with the electron beam at location **214b** is involved, i.e. impacting directly upon the substrate **210**, it may a priori be assumed that the object **230** must be moved correspondingly in order to allow a comparison with (subtraction from) an image captured with the electron beam directed towards the target material **220** as indicated at **214a** since X-ray radiation is generated at two different locations. FIG. 2 illustrates deflection of the electron beam **214a, b** between points separated a distance of BD on the target. The electron beam is shown as a solid arrow **214a** impacting a first point, and a dashed arrow **214b** impacting a second point on the target. As discussed, the first point is located in a region where the electron beam **214a** impacts the target layer **220** on the target, while the second point is located in a region where the electron beam **214b** does not impact the target layer **220** but rather directly impacts the substrate **210**. In order to obtain an image on the detector **240** at the same position for the two electron beam impact points, the sample or object **230** should be moved a distance OD, which may be calculated as

$$[00001] OD = BD \frac{ODD}{SDD}$$

where ODD is the distance from the sample to the detector (object detector distance) and SDD is the distance from the target to the detector (source-detector distance). The distance that the sample should be moved between image acquisitions may thus be calculated, by considering congruent triangles, to be the displacement of the electron beam on the target scaled by the ratio between the distance between the sample and the detector and the distance between the source and the detector. For many practical applications the distance between the source and the sample, corresponding to SDD-ODD as shown in FIG. 2, is small compared to the distance SDD from the source to the detector, which means that $SDD \approx ODD$, and the sample may thus be moved a distance OD substantially equal to the displacement BD of the electron beam without this having a noticeable difference on the final result. As an example, the distance ODD between the sample and the detector may be 100 times the distance (SDD-ODD) between the source and the sample. In such case the sample should in principle be moved 99% of the electron beam displacement. If beam displacement is of the order of 10 μm then the sample should be moved about 9.9 μm . The 0.1 μm

difference may be neglected especially since a typical pixel resolution on the detector is in the range of 100 μm . Thus, considering the magnification of about 100 in this example, the 0.1 μm error would only correspond to one tenth of a pixel. Similar arguments may be applied when considering thicker samples. The two sides of a thick sample are at different distances from the detector and motion should thus in principle be scaled differently for the two sides, but in practice an average scaling may be used without noticeably distorting the image.

[0035] FIG. 3 illustrates a method for X-ray imaging using an X-ray source comprising an electron source configured to provide an electron beam and a target configured to produce X-ray radiation upon impact by said electron beam. The method comprises directing 401 said electron beam to a first position on said target; recording 402, using a detector, a first X-ray image while directing said electron beam to said first position; directing 403 said electron beam to a second position; recording 404, using said detector, a second X-ray image while directing said electron beam to said second position; and generating 405 a difference image between the first image and the second image. As discussed above, for both the first and the second position of the electron beam, the X-ray radiation reaching the detector will comprise X-ray radiation generated by interaction between the electron beam and the beam-limiting element (the aperture), and any features or noise in the captured images can thus be reduced by generating such difference image between the two captured images.

[0036] To further reduce the effects caused by the secondary radiation, a scale factor may be determined from the respective exposure times used when capturing the first and the second images, and then scaling pixel values of the first and/or the second image before generating the difference image so that a difference in exposure time between the first and the second image may be compensated for.

[0037] To further improve the image quality of the final difference image, pixel values of the second image that are below a predetermined threshold may be set to zero before the difference image is generated.

[0038] Preferably, the method involves moving an object to be imaged so that a position of an image of the object on the detector is substantially the same for the first and the second images.

[0039] Depending on the circumstances, it may also be preferred to align the first and second images with each other using (per se known) image processing before generating the difference image.

[0040] The second position may correspond to a position on the target at which the electron beam impacts directly upon the target substrate. In such case, the method may further comprise directing the electron beam to a third position and capturing a third image, wherein the third position corresponds to the electron beam dump which is arranged so that substantially no X-ray radiation created by interaction between the electron beam and the electron beam dump is emitted from the X-ray source, at least no X-ray radiation that reaches the detector. A fourth image may then be created by subtracting pixel values of the third image from corresponding pixel values of the second image. The resulting pixel values of the fourth image can then be scaled by a scale factor, and the scaled fourth image as well as the third image can be subtracted from the first (main) image. In this way, secondary emission generated both at the aperture and in the target substrate is compensated for.

[0041] Arrangements, sources, and methods according to the invention may be used for different types of X-ray imaging such as X-ray microscopy, radiography, fluoroscopy, or CT scanning.

Claims

1-12. (canceled)

13. An X-ray imaging system, comprising: an X-ray source; a sample position; a detector arranged to detect X-ray radiation downstream of said sample position; wherein said X-ray source

comprises: an electron source arranged to provide an electron beam; a target arranged to produce X-ray radiation upon impact by said electron beam, the target comprising a substrate and a target layer at least partly covering said substrate, wherein said target layer is arranged to produce X-ray radiation upon impact by said electron beam; a deflector arranged to deflect the electron beam for directing the electron beam to a first position on said target layer and a second position selected from a position on said target at which the electron beam impacts directly upon the substrate and a position on an electron beam dump arranged so that substantially no X-ray radiation created by interaction between the electron beam and the electron beam dump reaches the detector; a controller arranged to record, using said detector, a first image with the electron beam directed to said first position, and a second image with the electron beam directed to said second position, and generate a difference image between the first image and the second image.

14. The X-ray imaging system of claim 13, further comprising a manipulator for moving the X-ray source and the sample position in relation to each other.

15. The X-ray imaging system of claim 13, further comprising a beam-limiting element arranged between the electron source and the target.

16. The X-ray imaging system of claim 15, wherein X-ray radiation generated from interaction between the electron beam and the beam-limiting element reaches the detector when the electron beam is directed to the first position and the second position.

17. An X-ray imaging system, comprising: an X-ray source; a sample position; a detector arranged to detect X-ray radiation downstream of said sample position; wherein said X-ray source comprises: an electron source arranged to provide an electron beam; a target arranged to produce X-ray radiation upon impact by said electron beam, the target comprising a substrate and a target layer at least partly covering said substrate, wherein said target layer is arranged to produce X-ray radiation upon impact by said electron beam; an actuator arranged to move the target in relation to the electron beam for directing the electron beam to a first position on said target layer and a second position selected from a position on said target at which the electron beam impacts directly upon the substrate and a position on an electron beam dump arranged so that substantially no X-ray radiation created by interaction between the electron beam and the electron beam dump reaches the detector; a controller arranged to: record, using said detector, a first image with the electron beam directed to said first position, and a second image with the electron beam directed to said second position, and generate a difference image between the first image and the second image.

18. The X-ray imaging system of claim 17, further comprising a manipulator for moving the X-ray source and the sample position in relation to each other.

19. The X-ray imaging system of claim 17, further comprising a beam-limiting element arranged between the electron source and the target.

20. The X-ray imaging system of claim 19, wherein X-ray radiation generated from interaction between the electron beam and the beam-limiting element reaches the detector when the electron beam is directed to the first position and the second position.

21. A method for X-ray imaging using an X-ray source comprising an electron source configured to provide an electron beam, and a target comprising a substrate and a target layer at least partly covering the substrate, wherein the target layer is arranged to produce X-ray radiation upon impact by the electron beam, the method comprising: directing said electron beam to a first position on said target layer; recording, using a detector, a first X-ray image while directing said electron beam to said first position; directing said electron beam to a second position; recording, using said detector, a second X-ray image while directing said electron beam to said second position; and generating a first difference image between the first X-ray image and the second X-ray image; wherein, the second position is selected from a position on said target at which the electron beam impacts directly upon the substrate and a position on an electron beam dump arranged so that substantially no X-ray radiation created by interaction between the electron beam and the electron beam dump reaches the detector.

- 22.** The method of claim 21, further comprising determining a scale factor from exposure times of the first X-ray image and the second X-ray image, respectively; and scaling pixel values of at least one of the first X-ray image and the second X-ray image before generating the first difference image.
- 23.** The method of claim 21, further comprising setting pixel values of the second X-ray image that are below a predetermined threshold to zero, and thereafter performing the step of generating the first difference image.
- 24.** The method of claim 21, further comprising: moving an object to be imaged so that a position of an image of the object on the detector is substantially the same for said first X-ray image and said second X-ray image.
- 25.** The method of claim 21, further comprising: aligning the first X-ray image and the second X-ray image before generating the first difference image.
- 26.** The method of claim 21, further comprising directing said electron beam to a third position and recording a third X-ray image, wherein said first position corresponds to a position on the target configured for X-ray production; said second position corresponds to an electron beam dump arranged so that substantially no X-ray radiation created by interaction between the electron beam and the electron beam dump reaches the detector; said third position corresponds to a position on the target not configured for X-ray production; generating a second difference image between the third X-ray image and the second X-ray image; scaling pixel values of the second difference image by a predetermined scale factor; and generating a third difference image between the first difference image and the scaled second difference image.
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